Two-length builders

The mathematics in one view

With unlimited copies of two positive integer rod lengths a and b, a target n is reachable exactly when n = ax + by for nonnegative integers x, y. Rod order does not matter; cutting, overlaps, and negative rods are forbidden.

Finite evidence can certify infinitely many targets. If a consecutive integers N, ..., N+a-1 are reachable, then every integer at least N is reachable: repeatedly add an a-rod to these a starting builds.

Exact boundary. If gcd(a,b)=1 and a,b>1, the largest impossible target is ab-a-b, so every target at least (a-1)(b-1) works. If gcd(a,b)>1, every target not divisible by that gcd is impossible, giving infinitely many gaps. A 1-rod makes every nonnegative target possible, so there is no largest positive gap in that case.

What each group is doing

K-1: Investigate small builds, distinguish rearranging from a new combination, and physically continue a run. Counting and alignment to 18 are enough; adults can record. A repeated construction is a genuine explanation without the formula.

Grades 2-3: Make exhaustive small lists, then find a short certificate for no later gaps. Addition to 28 and willingness to organize cases help. The shared-factor example refutes an overbroad conjecture.

Grades 4-5: Explain completeness, compare last gaps, conjecture the formula, and justify eventual coverage for a chosen coprime pair. Multiplication and remainders support the last questions; the general formula proof is adult backup.

Materials and preparation

For the current eleven children, use five pair-kits, one for each pair at the K-1 and 2-3 tables and one for the 4-5 trio: ten copies each of rods 2, 3, 4, 5 and 7 at 1 cm per unit (Cuisenaire scale), a few 6-rods or 6 cm paper strips for grades 4-5 P5, plus graph-paper strips and scissors for invented lengths. Existing measured rods or cut paper strips both work; the rod sheet, week-29-materials.pdf, prints one pair-kit with three 6-rods on one sheet of card stock. Every printed strip and ruler is 1 cm per unit, so print at 100%; the K-1 strips are outlines the rods must fill. Use the printed ruler scale when placing rods directly on a worksheet; otherwise build beside it on matching graph paper. Give each pair a pencil and a few target cards. Reuse rods between targets; large targets can be represented by counts.

Print only the starting pages and keep later problems available. The aim is a convincing construction or obstruction, not completion of all three packets.

Launch and a flexible hour

0-5, after the opening run-around: Launch. Let pairs handle rods, then build 7 from 3 and 4 on a printed ruler, such as the one in Grades 2-3 P1: start the train at 0 and read 7 at its end, as in the shared student example. Show that reversing the rods reaches the same target and is the same combination; a gap or overhang is not a build.

5-20: K P1-3, 2-3 P1-3, or 4-5 P1-2. **20-38:** Stay with the investigation or offer K P4-6, 2-3 P4-6, or upper P3-8; a pair that needs to move can carry one successful target and one doubtful target to the other pair at its table. **38 to the end:** Share one claim and the build or obstruction that establishes it. Leave the general formula as a conjecture if proof is not yet within reach. Minutes count from the launch; the tables have about 35-40 minutes.

K-1 answer key

P1. Of the listed targets, 6=3+3, 8=4+4, 9=3+3+3, 10=3+3+4 work. The targets 1, 2, 5 do not. One rod is too long for 1 or 2; for 5, one rod is too short and two rods total at least 6.

P2. For 12: four 3-rods or three 4-rods. For 15: five 3-rods, or one 3-rod and three 4-rods. These are all: after choosing the number of 4-rods, the remainder must be a multiple of 3. The third strip under each target stays empty.

P3. With 3 and 5, 4 and 7 are impossible. The others work: 8=3+5, 9=3+3+3, 10=5+5, 11=3+3+5, 12=3+3+3+3. For 7, no 5 leaves a multiple of 3: the possible remainders are 7 and 2.

P4. Start with 6=3+3, 7=3+4, 8=4+4; add 3 to reach 9,10,11, then 12,13,14, then 15,16,17, then 18. These three chains keep going forever and cover every integer at least 6.

P5. Exactly 2,4,6,8,10,12 work. Both rods are made of whole pairs of units; joining them always leaves an even length. Every positive even length is made from 2-rods.

P6. Exactly {2,3}, {2,5}, {3,4} cover every target 6-12. For {2,3}, use 2-rods for even targets and one 3 plus 2-rods for odd targets. For {2,5}, do the same with one 5 for odd targets. The failures are {2,4}: 7,9,11; {3,5}: 7; {4,5}: 6,7,11. This checks all six distinct pairs.

Hints, only when needed

For an “all ways” task, ask what happens with zero, one, two, ... of the longer rod. For an infinite claim, put three adjacent 3/4 builds side by side and ask whether each can grow. For an impossibility claim, ask for the full short list of possible longer-rod counts.

Grades 2-3 answer key

P1. In 1-16, only 1,2,5 fail. The certificate 6,7,8 plus added 3-rods covers all later targets.

P2. Writing (number of 3-rods, number of 4-rods): 12 has (4,0),(0,3); 16 has (4,1),(0,4); 24 has (8,0),(4,3),(0,6). The number of 4-rods has one required remainder modulo 3, so stepping by three 4-rods lists every possibility.

P3. The impossible lengths in 1-20 are 1,2,4,7. The last is 7: 8=3+5, 9=3+3+3, 10=5+5, followed by added 3-rods, settle all later targets.

P4. Yes to both. Minimal consecutive certificates are 6,7,8 for 3/4 and 8,9,10 for 3/5: add 3-rods to each, so the gaps stop at 5 and 7. Any other finite collection that demonstrably covers every remainder and starts beyond the last gap is acceptable, such as 17,18,19 or 21,22,23.

P5. Last gap 17. A certificate is 18=4+7+7, 19=4+4+4+7, 20=4+4+4+4+4, 21=7+7+7; then add 4-rods. For 17, using zero, one or two 7-rods leaves 17,10 or 3; none is divisible by 4.

P6. The claim is false: 2/4 misses every odd length. Of the ten pairs from 2, 3, 4, 5 and 7 it is the only one with gaps forever; the other nine share no divisor larger than 1. The pair 4/7 does eventually have no gaps, as P5 proves. One counterexample is enough to refute “every pair.”

Grades 4-5 answer key, P1-6

P1. Exactly 1,2,5. Check these small failures and use the three-build certificate above; a long tested list alone does not prove completeness.

P2. Last gaps: 3/5 → 7; 4/5 → 11. Certificates: 8,9,10 as above; 12=4+4+4, 13=4+4+5, 14=4+5+5, 15=5+5+5. For gap 11 with 4/5, zero, one, two 5-rods leave 11,6,1, none divisible by 4.

P3-4. All twelve targets 18-29 work: use the four builds 18=4+7+7, 19=4+4+4+7, 20=4+4+4+4+4, 21=7+7+7 and add 4 once or twice. Target 17 fails: zero, one or two 7-rods leave 17, 10 or 3, none divisible by 4.

P5. 2/4, 3/6 and 4/6 have infinitely many gaps (odd targets; nonmultiples of 3; odd targets and 2). The pairs 3/5 and 4/7 have only finitely many. The dividing line is a common divisor greater than 1; the coprime direction needs the argument on the next page.

P6. All ways for 24 are (8,0),(4,3),(0,6). Replace four 3-rods by three 4-rods, or reverse that replacement. Equality of two builds gives $3\Delta x=-4\Delta y$, forcing changes to be multiples of this exchange.

Grades 4-5 answer key, P7-8

P7. Last gap for 5/7 is 23. It is impossible because zero to three 7-rods leave 23,16,9,2, none divisible by 5. A certificate is 24=5+5+7+7, 25=5×5, 26=5+7+7+7, 27=5+5+5+5+7, 28=7×4. Adding 5 covers all later targets. The earlier last gaps 5,7,11,17 and this 23 fit ab-a-b. These examples support a conjecture, not a proof of the formula.

P8. Many choices work. Example: 2/5 has builds 4 and 5, so all targets at least 4 work. In general, assume a,b are positive and coprime. The a numbers 0,b,2b,...,(a-1)b have different remainders modulo a: equal remainders would make a divide (j-k)b and hence j-k, impossible for distinct indices in 0,...,a-1. Each remainder therefore has a reachable starting value; add a-rods within each remainder class. This proves eventual coverage for any invented pair. If a=1, the conclusion is immediate.

Why the exact last-gap formula holds

For a,b>1 coprime, set F=ab-a-b. If F=ax+by with x,y≥0, then a(x+1)+b(y+1)=ab. Reducing modulo b shows b divides x+1. Since x+1>0, the first term is at least ab, and the second term is positive: contradiction.

For N≥F+1, choose j in {0,...,a-1} so that jb and N have the same remainder modulo a. Then N-jb is a multiple of a and N-jb≥1-a. A multiple of a strictly greater than -a must be nonnegative, so N=ai+jb with i≥0. Thus F is impossible and every larger integer is possible.

For children, a small complete certificate is usually a better endpoint than this symbolic proof. Ask a child who says “it always works” to identify whether they have tested a range, found a repeatable construction, or proved the general statement.

Optional extension and continuity

If a group owns the coverage idea, ask for two rod pairs with the same last gap and a certificate for each: 2/7 and 3/4 both have last gap 5. This tests whether the boundary determines the pieces. Stop before turning the session into a large search table.

This is a new rod-reachability investigation in the current library, rather than a claim that divisibility itself is new to returning children. All grade adaptations here are unpiloted.

Primary source

Dilip Barman, Chapel Hill Math Circle, “The Frobenius Coin Problem,” Beginners’ Group (grades 1-3), 18 January 2025, pp. 1-4. Direct precedent for the coin-reachability question and a finite pattern of builds establishing eventual coverage. The rod adaptation, current tasks and proofs above were checked separately. Open the original handout.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

Keep one adult at each of the three tables and let partners alternate building and recording. On return, K-1 can grow a gap-free run (P4), contrast even-only rods (P5), or choose a pair (P6). Grades 2-3 can build finite coverage certificates and compare 4/7 with 2/4 (P4-6). Grades 4-5 can pursue rod exchanges, the last-gap prediction or the coprime coverage argument (P5-8). Keep a few successful adjacent targets visible instead of reconstructing a large table; the formulas are further explanations, not an introductory requirement.